



Master 2 Internship report

Dynamical properties of the Hopfield model through
bitwise arithmetic

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Abstract

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1 Introduction

1.1 Presentation of the host laboratories

This internship was conducted as part of my Master 2 at the Magistère of Fundamental Physics of Paris-Saclay Université, which I completed at Aix-Marseille Université in the “Physique Fondamentale et Applications” track. It was carried out at the “Physique des Cellules et Cancer” (PCC, UMR 168) laboratory of the Institut Curie as well as at the Gulliver laboratory (UMR 7083) of the “École Supérieure de Physique et de Chimie Industrielles” (ESPCI) in Paris, and was supervised by Michele Castellana (Institut Curie) and David Lacoste (ESPCI). Although the PCC unit is a leading biological physics laboratory, part of the Institut Curie – a pioneer institution in cancer research and treatment – it hosts a renowned theory team (of which Michele Castellana is a member) that is sometimes interested in physics-driven topics such as the one investigated in this report. The Gulliver laboratory hosts both experimentalists and theoreticians (among whom David Lacoste), working at the frontier between physical, chemical and biological topics.

1.2 Subject

In October 2024, John J. Hopfield was awarded the Nobel Prize in Physics alongside Geoffrey Hinton for their pioneering work on artificial neural networks [1]. Their work laid the foundations for further development and understanding of neural networks, whether they are organic or artificial, and eventually gave rise to Artificial Intelligence (AI).

Hopfield worked on an artificial neural network, originally invented by Shun’ichi Amari in 1972 [2]. He greatly contributed to the analysis, understanding and popularization of the network in his landmark paper *Neural networks and physical systems with emergent collective computational abilities* published in 1982 [3], and the network was eventually named after him. The model, described in more details in subsection 5.1, is similar to the Ising and the Spin Glass networks: it is composed of a set of spins – called neurons in the case of the Hopfield network – $-S_i \in \{-1, 1\}$ interacting with one another. In the same way as the two preceding models, the Hopfield model is energy-based, that is it will tend to minimize its energy over time. What makes this model different from the other two is the way the interactions between neurons are defined: as for the Ising and Spin Glass models they remain static over time but depend on the set of patterns to be stored in the network, making the latter attractors – minima of energy – of the model.

The Hopfield model provided the first simple yet powerful framework for understanding networks of interconnected neurons. It has become a reference in statistical physics on networks and its properties at equilibrium have been extensively studied and are now well known, whether it is its statistical mechanics [4–6] or its phase diagram and transitions [7]. However, much less is known about the dynamics that lead to those well-known equilibrium properties.

1.3 Objectives

The main objective of this internship is to efficiently simulate the Metropolis dynamics on the Hopfield network. To this end, we use a bitwise formalism that allows us to simulate several replicas of the system in parallel, increasing the efficiency of the exploration of the model’s energy landscape. Our objectives in this internship are thus twofold: first, to validate the bitwise implementation and quantify its computational gain in yield over the standard Metropolis algorithm; second, to leverage this implementation to study the dynamics of the Hopfield network. To validate the implementation, we begin with the Ising model, which serves as a natural benchmark due to its simplicity and well-characterized phase transition. Once this implementation has been proven sane and efficient on the Ising network, we increase the complexity by implementing it on the Spin Glass network, testing the gain in yield again.

Once the bitwise arithmetic has been validated on both models, we implement it on the Hopfield network in order to analyze its dynamics.

2 Implementation of the algorithm

We start by presenting the broad implementation of the Metropolis algorithm, before showing how we adapt it to each specific model.

2.1 The Metropolis algorithm

In order to make the models evolve in time, which is necessary to observe their dynamics, we use the well-known metropolis algorithm [8]. For models with a defined Hamiltonian, the Metropolis algorithm proceeds as follows:

1. Start from an initial configuration $\{S_i\}$.
2. Repeat N times:
 - a. Pick a random spin S_k and compute the energy change ΔE_k upon flipping it.
 - b. If $\Delta E_k \leq 0$, accept the flip.
 - c. Else, draw a random number $r \sim \mathcal{U}([0, 1])$ and accept the flip if $r \leq e^{-\beta \Delta E_k}$.
3. Repeat Step 2 for N_{sweeps} sweeps.

Where N is the number of spins, β is the inverse temperature, and N_{sweeps} is the number of sweeps, where one sweep consists of N successive single-spin flip attempts.

This algorithm however becomes computationally demanding when one attempts to sample a large number of configurations. This situation occurs, for example, when looking for rare samples in the dynamics, or sampling many different initial conditions – both of which we will try to explore during this internship. We thus need to find way of optimizing the implementation of the algorithm.

2.2 First optimization: the bitwise arithmetic

The first optimization in the implementation of the Metropolis algorithm is the use of bitwise arithmetic. The main idea is to leverage the boolean representation as stored in our computers. The typical object that we manipulate is called **Bits**: it is a collection of $n_{\text{bits}} = 64$ boolean variables. In that way, we have the possibility to simulate 64 evolutions of the same variable in parallel.

$$w = \underbrace{\boxed{b_0} \boxed{b_1} \boxed{b_2} \cdots \boxed{b_{62}} \boxed{b_{63}}}_{64 \text{ bits} \leftrightarrow 64 \text{ replicas}}$$

Figure 1: A **Bits** w encoding 64 parallel replicas. Each bit b_i carries the state of replica i .

This structure is very convenient to represent binary variable such as the spins of the system. The boolean representation $b_i \in \{0, 1\}$ of a spin $s_i \in \{-1, +1\}$ is given by $b_i = (s_i + 1)/2$. In that way, we are able to implement several replicas of the full spin system in a set of **Bits** $\{S_i\}$, each encoding the replicas of the associated spin, as shown in Figure 2.

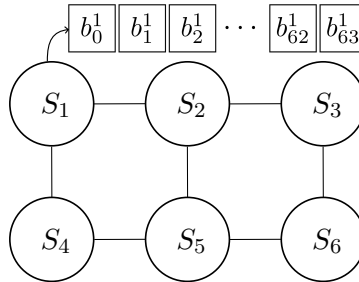


Figure 2: Bitwise implementation of a spin system. Each replica of the system is obtained by selecting the corresponding replica in each spin.

The limitation of n_{bits} to 64 is directly related to the architecture of modern CPUs [] [I need some sources here](#). This limit may in principle be extended up to 512 bits using SIMD instructions, but that requires the use of new object type in our code. This will be left for further discussion.

$$S_1 = \boxed{b_0^1} \boxed{b_1^1} \boxed{b_2^1} \cdots \boxed{b_{62}^1} \boxed{b_{63}^1}$$

2.3 Second Optimization: Shared random numbers

2.4 Structure of the code

3 The Ising model

3.1 The model

3.2 Bitwise Implementation

3.3 Phase diagram

3.4 Performance test

4 The Spinglass model

4.1 The model

4.2 Bitwise Implementation

4.3 Performance test

5 The Hopfield model

5.1 The model

5.2 Bitwise Implementation

5.2.1 Naive implementation

5.2.2 Optimized implementation

5.3 Capacity of the Hopfield network

5.4 Performance test

6 Conclusion

References

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